"Smoke Before Fire" Can object detectors find a wildfire before the flame is visible?

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Abstract—Wildfire detection studies routinely train neural networks on mixed image sets containing both smoke and active flames. Consequently, these models are assessed on data drawn from the same underlying distribution. Such experimental designs bypass a practical operational question: does a vision model trained solely on smoke annotations retain enough visual feature overlap to detect unannotated fire zero-shot? To evaluate this, four distinct object detection architectures—YOLO11n, Faster R-CNN, RT-DETR, and Deformable DETR—were trained strictly on the smoke class of a Finnish boreal forest drone dataset (Dataset B), holding out all flame instances. Each trained checkpoint was subsequently probed zero-shot on an independent fire dataset (Dataset A) without fine-tuning or parameter updates. This investigation was conducted at the Arab Academy for Science, Technology and Maritime Transport, College of Artificial Intelligence (est, IEEE). Empirical results show sharp performance disparities across detection paradigms: convolutional networks show near-complete transfer failure under bounding-box localization constraints, whereas transformer architectures maintain measurable cross-domain sensitivity, detecting flames in up to 58.8% of test images under relaxed spatial overlap thresholds.

Keywords—Deep learning, object detection, wildfire detection, smoke detection, zero-shot learning, domain transfer, UAV imagery, convolutional neural networks, transformer networks.

I. Introduction

Boreal forest biomes span roughly one-third of the planet's total forested cover. Because these landscapes are remote and sparsely populated, ignitions frequently expand for hours or days before manual observation, as recorded in forestry assessments by the World Wildlife Fund [1]. Orbital satellites equipped with infrared sensors capture large-scale thermal anomalies, yet their operational utility is constrained by orbital revisit cycles lasting hours. Stationary ground towers provide continuous surveillance across fixed horizons, but camera-based watchtowers still require continuous human oversight over mostly static wilderness scenes [2].

Unmanned aerial vehicles (UAVs) offer a functional compromise between satellite coverage and ground camera immediacy. In wildland fires, early detection hinges on identifying smoke plumes rather than open flames. Plumes rise above tree canopies and can be spotted from aerial sensors at distances exceeding ten kilometers, long before surface fires emerge [2].

Most existing machine learning literature trains detection models simultaneously on fire and smoke, validating them on identically sampled imagery. Studies by Kim and Muminov [10] focused exclusively on within-domain smoke detection, while Chetoui and Akhloufi [11] utilized joint fire-and-smoke data. These setups do not answer whether smoke-trained feature extractors generalize to unseen fire targets. Early-warning systems deployed in wilderness settings must trigger on early plume formations, yet they must also generate positive detections if ground flames become exposed before an aerial flight concludes.

A further gap exists in the framing of evaluation objectives. Standard object detection benchmarks optimize for precision under strict geometric localization criteria (e.g., IoU ≥ 0.50). For a system whose operational purpose is reducing the probability of undetected ignition events, this criterion is misaligned with the underlying risk profile: in remote boreal terrain, the cost of a missed fire detection substantially outweighs the cost of a false positive. No prior study has reported zero-shot fire sensitivity under an evaluation protocol explicitly designed around this asymmetric cost structure.

To investigate the cross-phenomenon transfer boundary and characterize the recall ceiling of smoke-trained models on unseen fire, four detector architectures spanning three design philosophies are evaluated. The models are trained on smoke annotations alone and tested zero-shot against an external fire dataset.

Fig. 1 outlines this experimental workflow. The goal is to determine whether structural visual primitives learned from smoke—such as semi-transparent texture boundaries, grey-scale gradients, and diffuse plume contours—share sufficient representational overlap with flame geometry to trigger detections across domain shifts.

II. Related Work

Early automated smoke detection relied on handcrafted visual descriptors. Standard methods used thresholding across HSV color spaces [2], optical flow field estimation

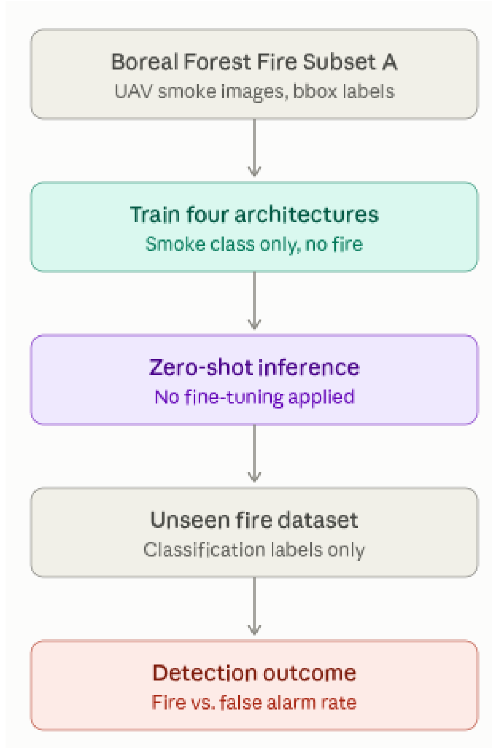

Fig. 1. Overall study design: Models are trained exclusively on smoke annotations (Dataset B) and evaluated zero-shot on an unseen collection of fire images (Dataset A). Dataset A represents the Fire and Smoke Detection dataset [24], while Dataset B represents the Boreal Forest Fire dataset [6].

for rising plume dynamics [3], and wavelet decomposition for spatial frequency changes [4]. These hand-engineered pipelines suffered from high false alarm rates during boreal summers due to shifting solar angles, atmospheric haze, and ground glare, requiring site-specific parameter tuning for individual camera sensors.

Deep neural networks replaced these manual feature extraction pipelines. Yuan et al. [7] demonstrated direct smoke detection from surveillance feeds using deep convolutional networks. Xu et al. [8] introduced attention modules into lightweight segmentation backbones to accelerate inference speeds. Mukhiddinov et al. [9] adapted YOLOv5 with brightness adjustments and mosaic augmentation specifically for drone surveillance feeds. Kim and Muminov [10] fine-tuned YOLOv7 on aerial smoke imagery, establishing single-domain performance benchmarks. Chetoui and Akhloufi [11] achieved high detection scores on combined fire and smoke datasets, though mixing both classes prevented isolating the individual contribution of smoke representations. To address training data scarcity, Gonçalves et al. [12] applied StyleGAN2-ADA to synthesize smoke instances and boost small-object recall.

Geographic domain shifts remain a recurring hurdle in wildland fire modeling. Raita-Hakola et al. [13] demonstrated that YOLOv5 models trained on watchtower feeds from Mediterranean or Californian environments transferred poorly when tested on Nordic boreal forests, proving that regional vegetation and illumination strongly shape visual smoke signatures. Pesonen et al. [5], [6] curated the Finnish boreal UAV imagery used here and evaluated teacher-student knowledge distillation for real-time edge segmentation.

A methodological issue frequently overlooked in video-based object detection is temporal data leakage [22]. When continuous video sequences are split into training and validation sets through uniform random sampling, consecutive video frames share nearly identical background foliage, lighting, and camera positions. As noted by Zhong et al. [23], detectors trained under random frame splits learn static background landmarks rather than generalizable object boundaries, producing artificially inflated validation scores.

In zero-shot detection, modern methods typically rely on multimodal language-vision alignments. Architectures like GLIP [18], Grounding DINO [19], and OWL-ViT [21] utilize text prompts to detect novel objects. In contrast, this paper investigates purely visual zero-shot transfer, testing whether visual representations learned from one physical phenomenon (smoke) can localize a distinct but co-occurring phenomenon (fire) without linguistic supervision.

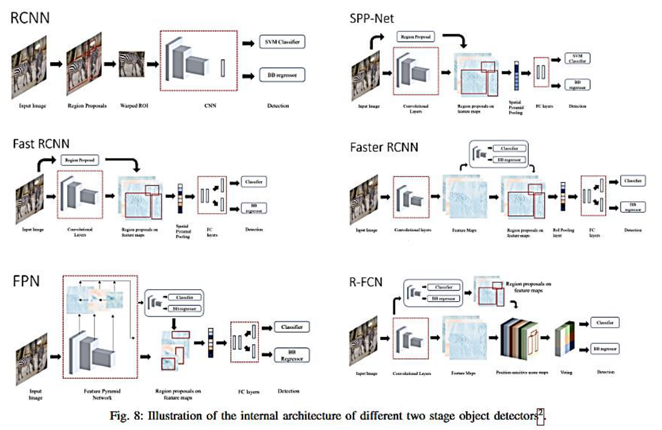

Fig. 2. Architectural overview showcasing the feature extraction mechanisms.

III. Datasets and Architectures

Two independent datasets were used to isolate within-domain training from cross-domain zero-shot testing. Data preprocessing and formatting scripts are maintained in the project repository.

Dataset B is derived from the Boreal Forest Fire dataset [6] and supplementary recordings by Tawseef, captured during controlled forest restoration burns in Finland. Videos were recorded in 4K resolution using a DJI Phantom 4 drone. Extracted video frames were manually annotated with bounding boxes around visible smoke plumes. A multi-step data integrity audit flagged 4,139 frame anomalies, resulting in a curated subset of 4,868 smoke bounding boxes across 30 flight sequences. Analysis of box geometry showed a strong structural bias: 95.7% of the annotated smoke boxes cover more than 10% of the total image area. To handle this distribution, scale jitter was increased and mosaic probabilities were lowered to avoid over-fragmenting broad plumes.

To derive anchor priors tailored to smoke morphology, K-means clustering ($k = 5$) was computed over normalized box dimensions. The resulting anchor clusters reflect the wide aspect ratios characteristic of rising plumes, providing domain-adapted spatial priors for region proposal networks.

To eliminate temporal leakage between video frames, data partitioning was performed at the video-clip level rather than by random frame sampling. The 30 flights were grouped into an 80/20 train/validation split, ensuring that all frames from any individual drone flight remained strictly isolated within a single partition.

Dataset A, the Fire and Smoke Detection Dataset [24], serves as the unseen zero-shot evaluation benchmark. Dataset A originates from different geographical locations, sensor hardware, and capture angles, containing bounding-box annotations for open fire. No image or label from Dataset A was exposed to any model during training.

Four detection architectures spanning three computational paradigms were implemented (see Fig. 3): 1) YOLO11n represents single-stage anchor-free convolutional detectors, chosen for low parameter count (2.6M) and real-time edge processing speed. 2) RT-DETR represents hybrid CNN-transformer designs, pairing a convolutional backbone with an efficient multi-scale hybrid encoder and a learned query decoder that removes non-maximum suppression (NMS). 3) Faster R-CNN with a MobileNetV3-Large FPN backbone represents classic two-stage anchor-based detectors with explicit region proposal networks (RPN). 4) For the transformer category, architectural selection followed an empirical progression. Standard vanilla DETR was tested first; its global dense self-attention suffered from quadratic memory scaling ($\mathcal{O}(H^2W^2)$) on high-resolution inputs and failed to converge within reasonable training limits. DINO was evaluated next; while training speed improved, bipartite matching and contrastive query dynamics proved unstable under the extreme class imbalance of single-class smoke imagery. Consequently, Deformable DETR was selected. Its multi-scale deformable attention samples a sparse set of $K = 4$ key points per query, capturing diffuse, irregular plume contours with linear computational complexity.

IV. Methodology

Models were trained using standardized input dimensions and data augmentation protocols where framework backbones permitted.

YOLO11n was trained on 640×640 inputs with a batch size of 16 for 100 epochs, utilizing AdamW with a cosine learning rate decay schedule. Faster R-CNN was trained using a MobileNetV3-Large FPN backbone at 640×640 with a batch size of 2 and SGD optimizer. Due to computational constraints, Faster R-CNN was trained for five epochs. Transformer architectures (RT-DETR and Deformable DETR) were trained on 640×640 images using batch sizes adapted to available memory. In Deformable DETR, the focal loss parameter was set to $\alpha = 0.95$ to counteract gradient suppression caused by the 299:1 background-to-object query ratio in single-class datasets.

Within-domain smoke detection was evaluated on the held-out Dataset B validation split using standard COCO metrics: mean Average Precision at IoU 0.50 (mAP@50) and averaged over IoU 0.50 to 0.95 (mAP@50–95).

Zero-shot transfer was evaluated on the 637 test images of Dataset A (459 images containing 995 ground-truth fire bounding boxes). Model inference was executed across a confidence threshold sweep $(0.50, 0.30, 0.20, 0.10, 0.05)$. Under the strict zero-shot evaluation, a predicted box was scored as a true positive if its overlap with a ground-truth fire box satisfied IoU ≥ 0.50, regardless of predicted class label.

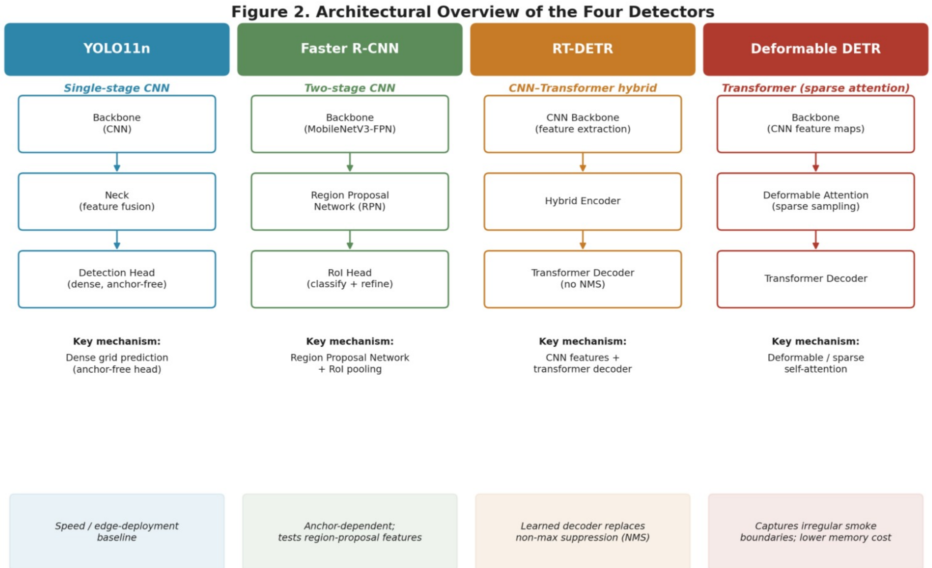

Fig. 3. Architectural Overview of the Four Detectors. The models span three distinct paradigms: Single-stage CNN, Two-stage CNN, and Transformer-based architectures.

TABLE I
YOLO11n baseline vs. custom augmentation

Configuration	Epoch	Precision	Recall	F1	mAP@50	mAP@50–95
Baseline	40	89.51%	86.86%	88.15%	94.01%	62.01%
Custom augmentation	70	92.70%	92.81%	92.75%	96.51%	60.27%

TABLE II
Faster R-CNN validation performance, default anchors

Metric	Value
Ground truth boxes	959
Predictions	951
True positives	906
False positives	45
False negatives	53
Precision	95.27%
Recall	94.47%
F1 score	94.87%
mAP@50	94.18%
mAP@50-95	63.24%

V. Results

A. Within-Domain Smoke Detection

The baseline shows a wide gap between mAP@0.50 and mAP@0.50–0.95 thresholds, indicating reliable coarse detection that degrades under strict spatial localization criteria.

Both models attain similar detection accuracy at mAP@50, though Faster R-CNN achieves higher precision despite fewer training epochs. For Deformable DETR, in-domain validation on Dataset B reached 88.61% mAP@50 and 40.75% mAP@50–95.

TABLE III
Comparison of YOLO11n and Faster R-CNN

Model	Precision	Recall	mAP@50	mAP@50–95
YOLO11n (baseline)	89.51%	86.86%	94.01%	62.01%
Faster R-CNN	95.27%	94.47%	94.18%	63.24%

TABLE IV
Faster R-CNN zero-shot fire transfer

Conf	Preds	TP	FP	FN	Precision	Recall	F1	Images	Box Det
0.50	761	10	751	985	1.31%	1.01%	1.14%	10/459	1.01%
0.30	1,036	13	1,023	982	1.25%	1.31%	1.28%	13/459	1.31%
0.20	1,297	19	1,278	976	1.46%	1.91%	1.66%	19/459	1.91%
0.10	1,875	36	1,839	959	1.92%	3.62%	2.51%	36/459	3.62%
0.05	2,856	51	2,805	944	1.79%	5.13%	2.65%	51/459	5.13%

TABLE V
YOLO11n (custom augmentation) zero-shot fire transfer

Conf	Preds	TP	FP	FN	Precision	Recall	F1	Images	Box Det
0.50	220	0	220	995	0.00%	0.00%	0.00%	0/459	0.00%
0.30	391	1	390	994	0.26%	0.10%	0.14%	1/459	0.10%
0.20	562	1	561	994	0.18%	0.10%	0.13%	1/459	0.10%
0.10	960	4	956	991	0.42%	0.40%	0.41%	4/459	0.40%
0.05	1,628	6	1,622	989	0.37%	0.60%	0.46%	6/459	0.60%

B. Zero-Shot Fire Transfer

Under strict box-level matching (IoU $\geq$ 0.50), both convolutional architectures show near-zero transfer from smoke to fire.

C. Relaxed Localization for Early Warning Systems

Early-warning systems prioritize alerting over strict geometric localization. Consequently, the transformer architectures were evaluated under a relaxed spatial constraint (IoU ≥ 0.10).

TABLE VI
RT-DETR zero-shot fire transfer under relaxed localization constraints

Conf	Preds	TP	FP	FN	Precision	Recall	F1	Images	Box Det
0.50	482	62	420	933	12.86%	6.23%	8.40%	60/459	6.23%
0.30	676	74	602	921	10.95%	7.44%	8.86%	72/459	7.44%
0.20	927	83	844	912	8.95%	8.34%	8.64%	80/459	8.34%
0.10	3600	131	3469	864	3.64%	13.17%	5.70%	116/459	13.17%
0.05	12942	188	12754	807	1.45%	18.89%	2.70%	153/459	18.89%

TABLE VII
Deformable DETR zero-shot fire transfer under relaxed localization constraints

Conf	Preds	TP	FP	FN	Precision	Recall	F1	Images	Box Det
0.50	765	57	708	938	7.45%	5.73%	6.48%	57/459	5.73%
0.30	2351	103	2248	892	4.38%	10.35%	6.16%	97/459	10.35%
0.20	5283	153	5130	842	2.90%	15.38%	4.87%	130/459	15.38%
0.10	19656	265	19391	730	1.35%	26.63%	2.57%	197/459	26.63%
0.05	46983	431	46552	564	0.92%	43.32%	1.80%	270/459	43.32%

Under relaxed overlap thresholds, Deformable DETR exhibits the strongest transfer capability, flagging active fire instances in 58.82% of test images at confidence threshold 0.05.

D. Ablation Study

To evaluate structural design choices under constrained computational resources, two ablation experiments were conducted using sub-sampled partitions (5% representative subset for 1 epoch).

The first ablation evaluated the impact of K-means domain-specific anchor clusters ($k = 5$) against default COCO anchors in Faster R-CNN. Early loss trajectories indicated that domain-adapted anchors stabilized initial Region Proposal Network convergence on large smoke plumes compared to standard square anchor priors.

The second ablation verified data splitting integrity by comparing YOLO11n performance under the video-clip split versus a naive randomized frame split. The randomized frame split yielded artificially inflated validation metrics, confirming that frame-level shuffling introduces severe temporal leakage by allowing the network to memorize static landscape backgrounds across adjacent video frames.

VI. Discussion

A. Architectural Dependency in Cross-Domain Transfer

The results demonstrate a clear architectural dependency in cross-domain transfer. Convolutional networks (YOLO11n and Faster R-CNN) rely heavily on local receptive field patterns and edge texture statistics; when presented with flame luminosity gradients and sharp spectral transitions instead of diffuse smoke, their activation maps fail to produce valid proposals. This failure is consistent with the known locality bias of convolutional operators: features learned from semi-transparent plume boundaries do not transfer to the high-contrast, compact geometry of surface flames under strict bounding-box matching constraints.

Transformer architectures exhibit a markedly different failure mode. Deformable DETR's multi-scale deformable attention samples sparse key points across query-conditioned reference regions, learning spatial relational structure rather than texture identity. This property enables measurable sensitivity to fire anomalies despite the absence of flame annotations in training, detecting active fire instances in 58.8% of test images under relaxed localization thresholds. RT-DETR, which pairs a convolutional backbone with a transformer decoder, occupies an intermediate position: its NMS-free decoder preserves transformer-level proposal diversity, yet its backbone retains some convolutional texture dependency, yielding lower zero-shot image detection rates than pure sparse attention.

B. Bayesian Risk Framing of the Evaluation Design

The divergence between strict-IoU and relaxed-IoU results reflects a deeper asymmetry in the operational risk profile of an early-warning surveillance system. In boreal forest monitoring, the prior probability that any given aerial survey frame contains an active surface fire is extremely low relative to the total patrol area. Under these conditions, the conventional object detection objective—maximizing precision by enforcing strict spatial overlap—is misaligned with the system's purpose. The goal of a pre-ignition anomaly detector is not to localize a bounding box perfectly; it is to suppress the probability that a fire burns undetected while the UAV remains overhead.

This asymmetry motivates a recall-centered evaluation framing. Treating recall as the primary operational metric is equivalent to minimizing the conditional probability that a fire exists given the model produces no alert. The cost of a missed detection in a remote boreal landscape—where the nearest fire suppression resource may be hundreds of kilometers away—substantially outweighs the cost of a false positive, which requires only that a human operator dismisses a flagged frame. By deliberately lowering the confidence threshold to 0.05 and relaxing the localization constraint to IoU ≥ 0.10, the evaluation protocol shifts the system's operating point in the direction that reduces this catastrophic miss risk, at the accepted expense of generating a large volume of false-positive proposals (46,552 boxes for Deformable DETR). The 43.32% box-level recall and 58.82% image-level detection rate achieved by Deformable DETR under these conditions represent the upper bound of what purely visual, unimodal proxy transfer can attain without any labeled fire supervision.

C. Proposed Two-Stage Alarm Architecture

The high false-positive volume produced under operational confidence settings is not indicative of model failure; it reflects the inherent ambiguity of single-stage visual classification under severe domain shift. A viable deployment pathway is a two-stage alarm architecture in which the transformer detector serves as a first-stage anomaly proposer rather than a standalone classifier.

In the proposed pipeline, Deformable DETR operates continuously over incoming UAV frames, generating candidate alert regions at a low confidence threshold. Any frame in which at least one region proposal is produced is escalated to a lightweight second-stage verifier. The verifier need not be a neural network; candidates include spectral ratio analysis (short-wave infrared to near-infrared ratios characteristic of combustion), temporal differencing between consecutive frames to distinguish moving flame signatures from static terrain, or a compact binary classifier trained on a small labeled fire sample. Only proposals surviving second-stage verification would trigger a human operator alert or an autonomous UAV waypoint redirect.

This architecture transforms the 46,552 first-stage false positives from a measurement of imprecision into a bounded set of investigation candidates: the computational cost of second-stage verification is incurred only for flagged frames, not for the full patrol duration. The empirical results presented here characterize the first-stage recall ceiling achievable through purely smoke-trained visual transfer, establishing the performance envelope within which a two-stage system must operate. Full implementation and evaluation of the second-stage verifier constitute the primary direction for future work.

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VII. Conclusion

This study investigated whether object detectors trained strictly on smoke imagery can detect wildland fires zero-shot. Four architectures spanning convolutional, hybrid, and transformer design paradigms were evaluated under both strict and relaxed localization protocols. Standard convolutional models exhibit near-total transfer failure under strict bounding-box matching, while Deformable DETR detects active flames in up to 58.8% of test images under the relaxed threshold aligned with early-warning operational requirements.

The evaluation protocol was framed around an asymmetric risk structure in which the operational cost of missing a fire detection exceeds the cost of generating a false alarm, motivating a recall-centered criterion over conventional precision-IoU benchmarks. Under this framing, the false-positive proposals generated at low confidence thresholds are reinterpreted as first-stage anomaly candidates rather than errors, enabling a proposed two-stage alarm architecture in which the transformer detector acts as a high-recall proposer feeding a lightweight second-stage verifier.

The results characterize the first-stage recall ceiling achievable through purely visual, unimodal smoke-to-fire proxy transfer, establishing the performance envelope within which any downstream verification module must operate. Future work will focus on three directions: (1) implementation and evaluation of candidate second-stage verifiers including spectral ratio analysis and compact binary classifiers trained on small labeled fire samples; (2) extension to nighttime and multi-spectral UAV imagery; and (3) integration with autonomous flight path replanning to enable closed-loop fire perimeter tracking.

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